ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 1 · PRACTICE D

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Laid out like the DHET first paper. Section A (20): multiple choice, true or false and choosing terms from a list · Section B (80): batteries for mobile robots, AGVs and AMRs, vacuum grippers for glass sheets, serial communication (I2C, SPI and UART) and data rates, delta robots and the kinematics of a two-link arm, ADC readings, PWM and optocouplers, and safety when testing a prototype. Formula sheet at the end.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B, and a formula sheet.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Use only a black or blue pen. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 A robot designed to work safely next to people without a safety fence is a …

A gantry robot. B collaborative robot (cobot).
C delta robot. D SCARA robot.

1.1.2 A vacuum gripper holds a part by means of …

A magnetism. B the friction of two fingers.
C a pressure difference. D glue.

1.1.3 AGV stands for …

A automated guided vehicle. B automatic gear valve.
C analogue gain voltage. D assisted ground vision.

1.1.4 The number of independent movements that a robot can make is its …

A payload. B reach.
C repeatability. D degrees of freedom.

1.1.5 Which 3D-printing process uses light to harden liquid resin layer by layer?

A FDM B SLA C CNC milling D Laser cutting

1.1.6 Which address is fixed in a device's network card by the manufacturer?

A IP address B MAC address C URL D Port number

1.1.7 A light curtain in a robot cell is used to …

A stop the robot when a person enters the danger zone. B light up the work cell.
C measure the colour of parts. D charge the robot's batteries.

1.1.8 Forward kinematics calculates …

A the joint angles needed to reach a target.
B the motor current.
C the cycle time.
D the end-effector position from known joint angles.

1.1.9 Which wireless technology is designed for short-range, low-power links between sensors and phones?

A 4G LTE
B Satellite
C Bluetooth Low Energy
D Microwave radio

1.1.10 Which device changes a controller's low-current signal into the large current that a motor needs?

A Sensor
B Encoder
C Motor driver
D Gyroscope

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 A cobot works with lower speed and force limits than a fenced industrial robot so that it can work safely next to people.

1.2.2 An absolute encoder loses its position when the power is switched off.

1.2.3 PLA is a common filament for FDM 3D printers.

1.2.4 When an emergency stop is released, the robot must start again automatically.

1.2.5 A wireless mesh network can keep working if one of its nodes fails.

(5)

1.3 Choose ONE term from the list below for each of the following descriptions, and write it next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

collaborative robot; MAC address; vacuum gripper; light curtain; AGV; cycle time; offline programming; servo motor; firmware; PLC; teach mode; slicer

1.3.1 Programming a robot in a computer simulation without stopping production

1.3.2 The time a robot takes to complete one full task

1.3.3 Software stored permanently in a device's memory that controls its hardware

1.3.4 A motor with built-in position feedback that turns to a commanded angle

1.3.5 A mobile robot that follows a fixed route (lines, magnetic tape or markers) to carry materials

(5)

[20]

TOTAL SECTION A: [20]

SECTION B

QUESTION 2

2.1 A mobile robot uses a 24 V, 20 Ah lithium-ion battery. The motors and electronics draw an average current of 8 A. Calculate the:

2.1.1 running time of the robot on one charge (ignore losses) (2)

2.1.2 energy stored in the battery in watt-hours (2)

2.2 Give TWO advantages and TWO disadvantages of lithium-ion batteries for mobile robots. (4)

2.3 Explain the difference between an AGV and an AMR (autonomous mobile robot). (3)

2.4 Name TWO ways in which mobile robots can charge their batteries without a person plugging them in. (2)

[13]

QUESTION 3

A glass factory wants a robot to lift flat glass sheets with a mass of 60 kg from a stack and place them on a cutting table. The design team chose a vacuum gripper with round suction cups.

3.1 Explain how a vacuum (suction) gripper holds the glass. (3)

3.2 Each suction cup has a diameter of 0,1 m, and the vacuum gives a pressure difference of 60 kPa.

3.2.1 Calculate the holding force of ONE suction cup. (3)

3.2.2 Calculate how many suction cups are needed if a safety factor of 2 is used. (3)

3.3 Name TWO other types of grippers and give ONE suitable use of EACH. (4)

[13]

QUESTION 4

4.1 Explain the difference between serial and parallel communication. (2)

4.2 For EACH of the following serial buses, state how many signal wires it uses and give ONE typical use in a robot: (6)

4.2.1 I2C

4.2.2 SPI

4.2.3 UART

4.3 A force sensor sends 2 000 readings every second, and each reading is 16 bits long. Calculate the data rate in kbit/s. (3)

4.4 Explain what a communication protocol is, and why two devices must use the same protocol. (2)

4.5 Give THREE possible causes of communication faults on a factory network. (3)

[16]

QUESTION 5

5.1 Describe a delta (parallel) robot and give ONE application of it. (3)

5.2 A two-link robot arm moves in a flat (X–Y) plane. Link 1 is 0,5 m long and link 2 is 0,3 m long. Joint 1 is at 30° from the X-axis and joint 2 is turned a further 60°. Calculate the X and Y position of the end of the arm. (6)

5.3 The robot accelerates a 4 kg part at 3 m/s². Calculate the force needed. (2)

5.4 Explain what a robot's work envelope is, and why it is important when a robot cell is planned. (3)

5.5 Give THREE advantages of using robots for welding. (3)

[17]

QUESTION 6

6.1 Explain the difference between a digital input and an analogue input, and give ONE sensor example of EACH. (4)

6.2 A 10-bit analogue-to-digital converter (ADC) with a 5 V reference gives a reading of 614. Calculate the input voltage. (3)

6.3 Explain what pulse-width modulation (PWM) is, and what a duty cycle of 25% means for the speed of a DC motor. (3)

6.4 Explain why an optocoupler is used between a microcontroller and a motor-driver circuit. (3)

[13]

QUESTION 7: PRACTICAL ROBOTICS

7.1 Give FOUR safety rules to follow when a new robot prototype is tested in the workshop. (4)

7.2 Explain the purpose of a user manual for a robot, and name TWO things that it must contain. (4)

[8]

TOTAL SECTION B: [80]

TOTAL: 100

END OF QUESTION PAPER

FORMULA SHEET

Battery running time: t = capacity (Ah) ÷ I	Battery energy: $E = V \times$ capacity (Wh)
Pressure force: $F = p \times A$; area of a circle: $A = \pi \times r^2$	Weight: $W = m \times g$ ($g = 9{,}8$ m/s²)
Force: $F = m \times a$	Data rate = readings per second × bits per reading
Two-link arm: $x = L_1 \cos \theta_1 + L_2 \cos (\theta_1 + \theta_2)$	$y = L_1 \sin \theta_1 + L_2 \sin (\theta_1 + \theta_2)$
ADC: V = reading × $V_{ref} \div (2^n - 1)$	Ohm's law: $V = I \times R$; power: $P = V \times I$